

Registered ABE and Adaptively-secure Broadcast Encryption from Succinct LWE

Jeffrey Champion, Yao-Ching Hsieh, David Wu

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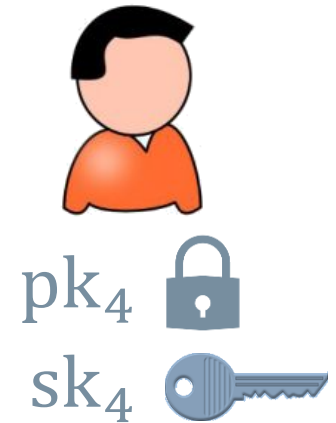
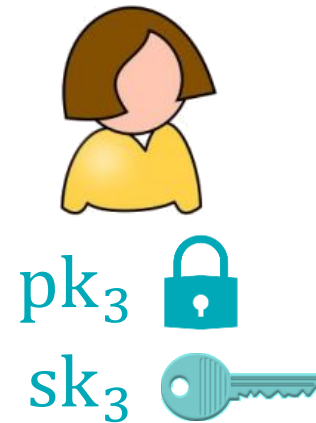
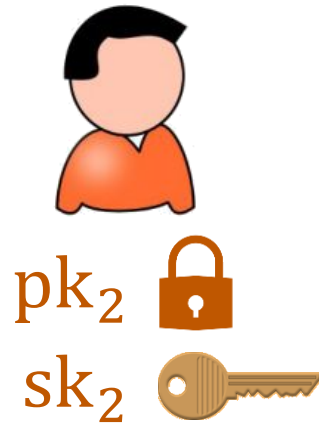
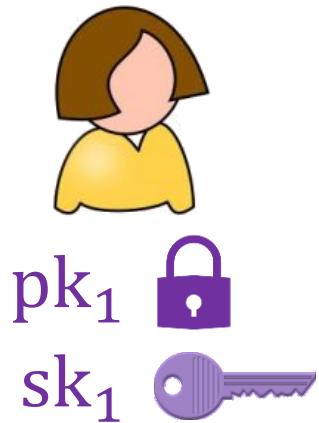
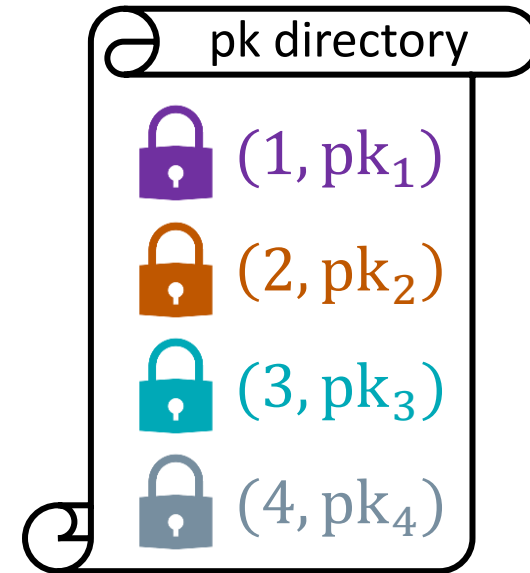
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Distributed Broadcast Encryption

[WQZD10, BZ14]



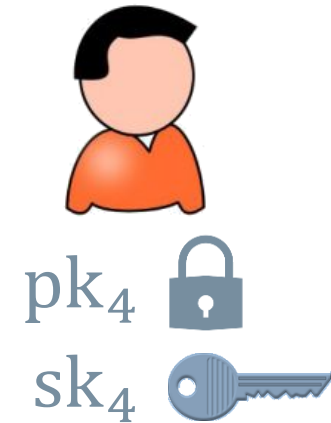
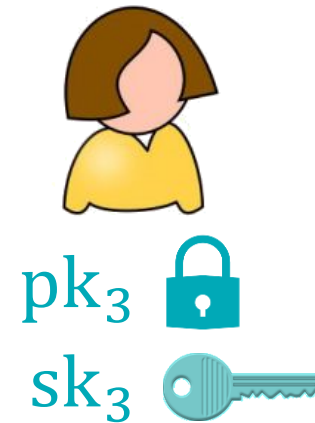
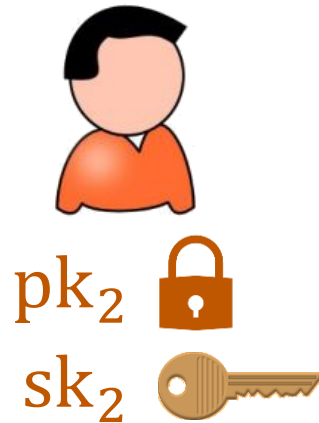
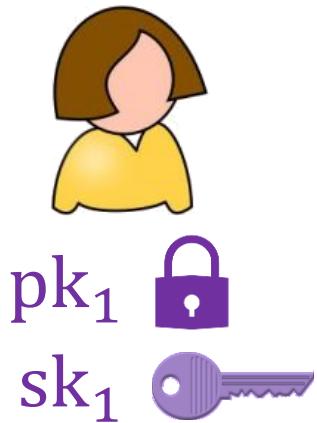
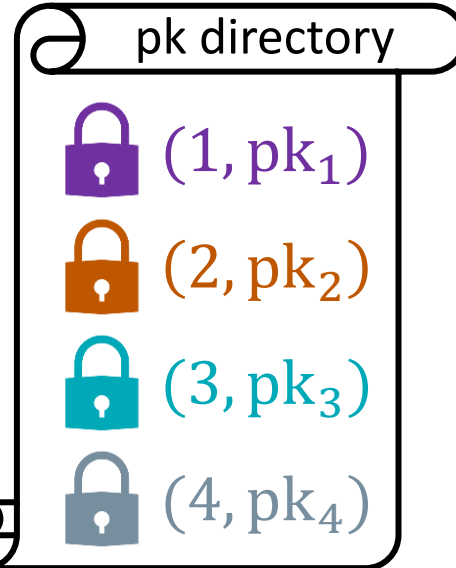
Distributed Broadcast Encryption

[WQZD10, BZ14]

message m

$S = \{1, 4\}$

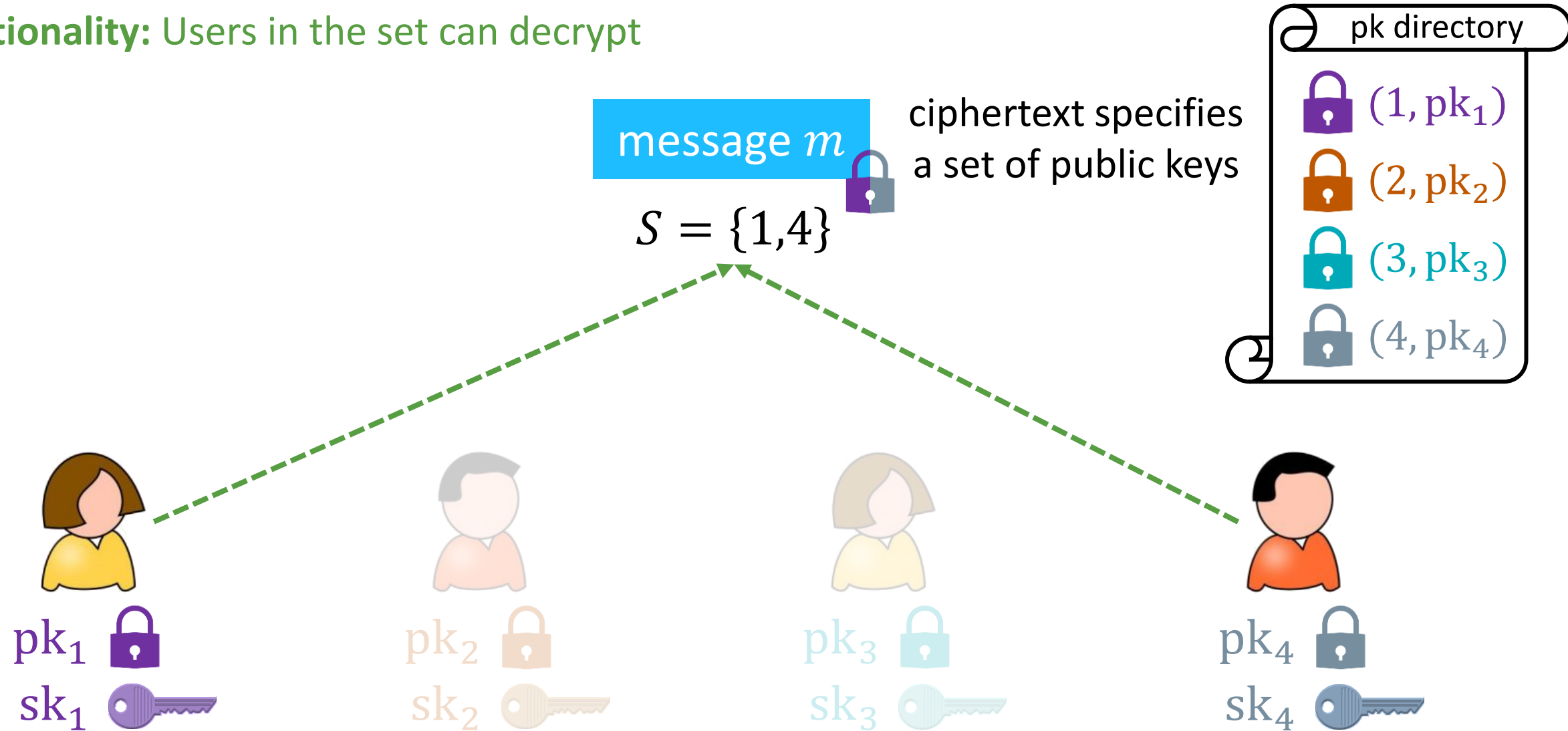
ciphertext specifies
a set of public keys



Distributed Broadcast Encryption

[WQZD10, BZ14]

Functionality: Users in the set can decrypt

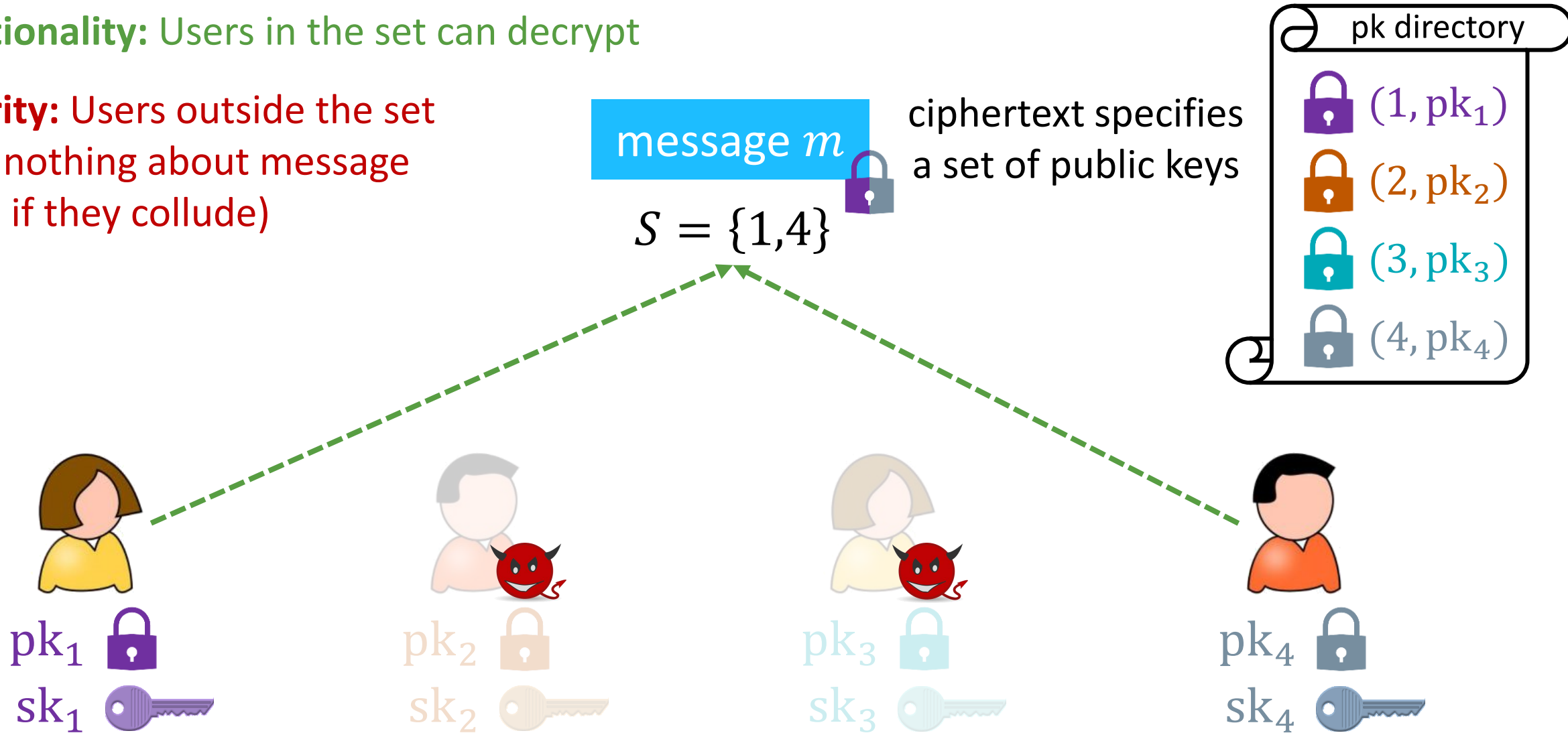


Distributed Broadcast Encryption

[WQZD10, BZ14]

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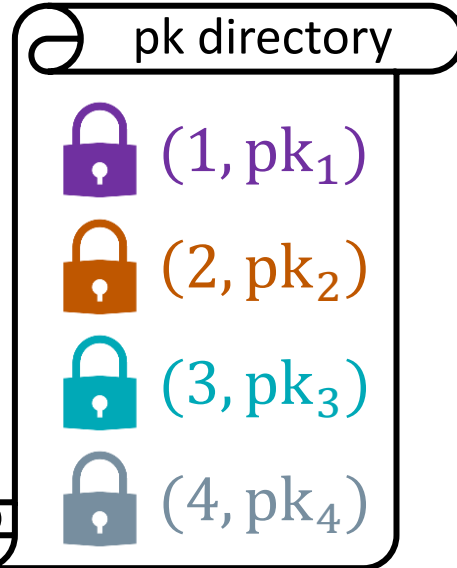
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Efficiency: $|ct| = |m| + \text{poly}(\lambda, \log|S|)$

message m

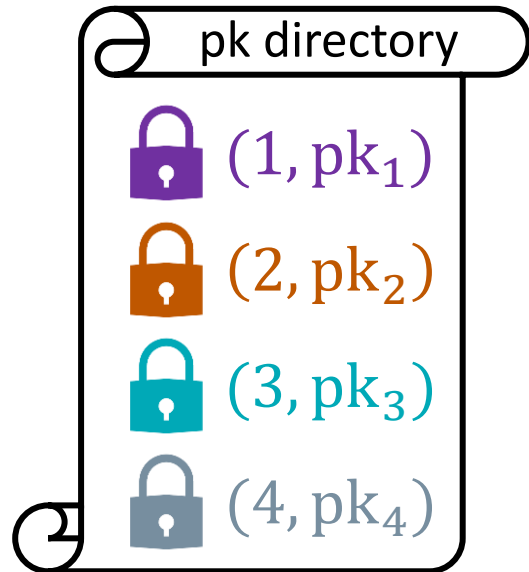
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Distributed Broadcast Encryption (DBE)

[WQZD10, BZ14]



public parameters

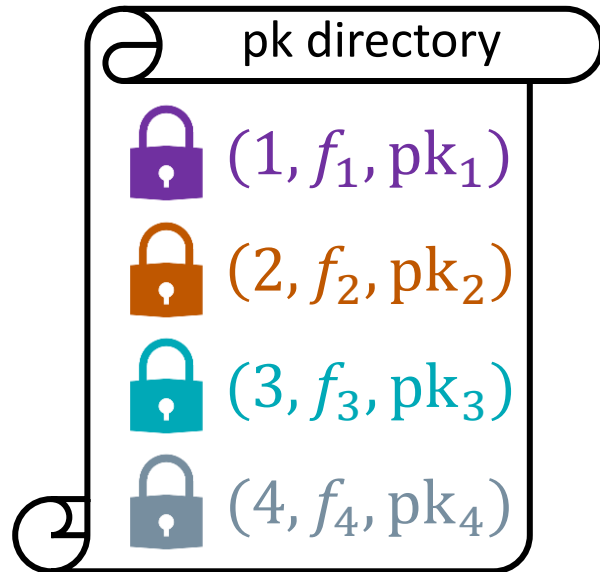
$$\text{KeyGen}(\text{pp}, i) \rightarrow (pk_i, sk_i)$$

$$\text{Encrypt}(\text{pp}, \{pk_i\}_{i \in S}, m) \rightarrow ct$$

$$\text{Decrypt}(\text{pp}, \{pk_i\}_{i \in S}, sk_i, ct) \rightarrow m$$

- Trustless version of traditional broadcast encryption [FN93]
- DBE implies traditional broadcast encryption with a long public key

Distributed Attribute-Based Encryption



public parameters

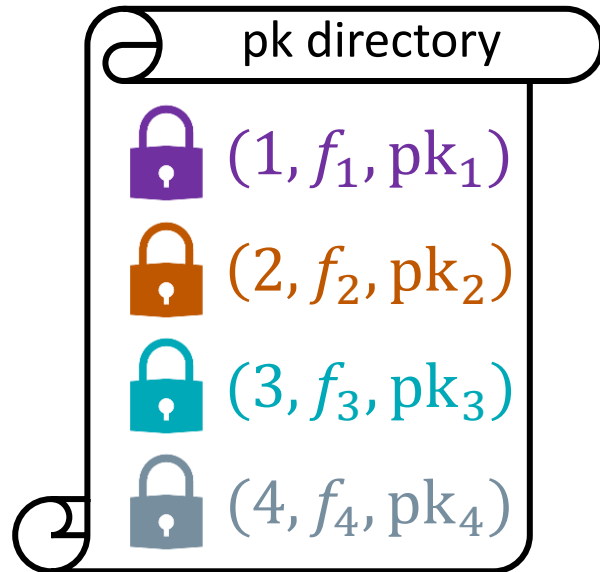
$\text{KeyGen}(\text{pp}, i, f_i) \rightarrow (\text{pk}_i, \text{sk}_i)$

$\text{Encrypt}(\text{pp}, \{\text{pk}_i\}_{i \in [N]}, x, m) \rightarrow \text{ct}$

size $\text{poly}(\lambda, \log N)$

$\text{Decrypt}(\text{pp}, \{\text{pk}_i\}_{i \in [N]}, \text{sk}_i, x, \text{ct}) \rightarrow m$

Distributed Attribute-Based Encryption



Functionality: Secret key for f_i recovers m if $f_i(x) = 0$

public parameters

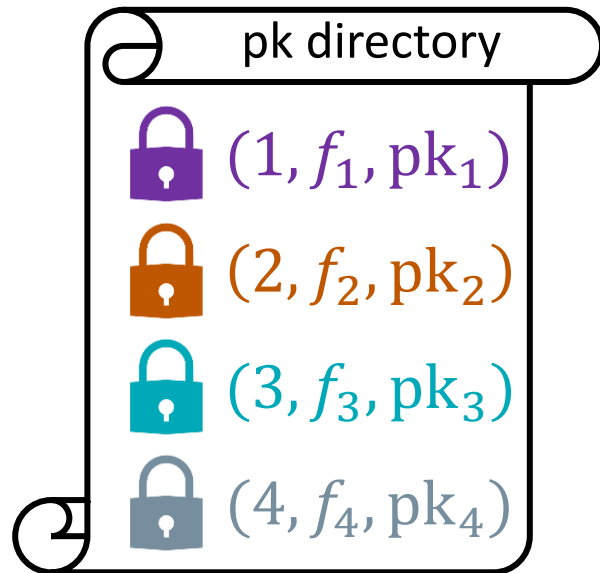
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Distributed Attribute-Based Encryption



Functionality: Secret key for f_i recovers m if $f_i(x) = 0$

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public parameters

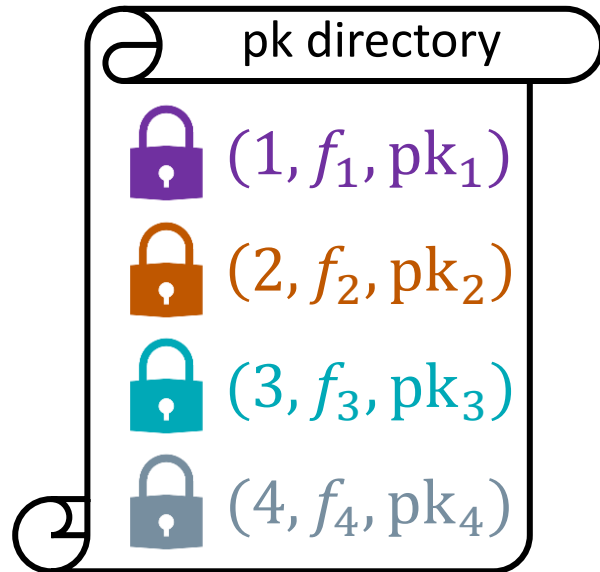
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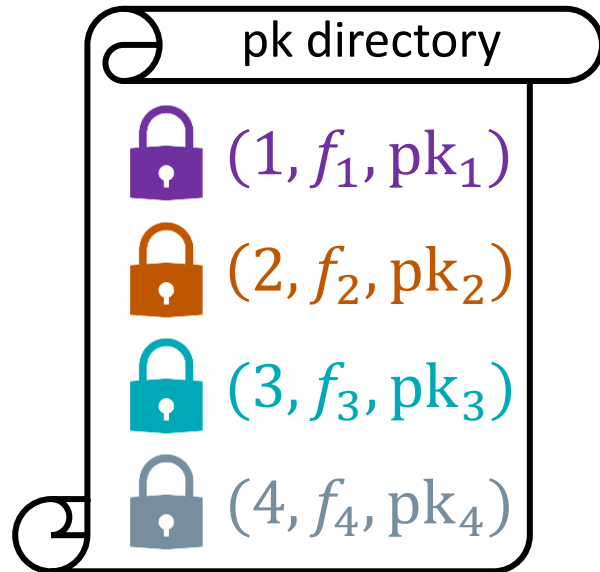
size $\text{poly}(\lambda, \log N)$

$\text{Decrypt}(\text{pp}, \{\text{pk}_i\}_{i \in [N]}, \text{sk}_i, x, \text{ct}) \rightarrow m$

User set is fixed but encryption/decryption time scales at least linearly with N !

Registered ABE

[GHMR18, HLWW23]



$\text{KeyGen}(\text{pp}, i, f_i) \rightarrow (\text{pk}_i, \text{sk}_i)$

size $\text{poly}(\lambda, \log N)$

$\text{Aggregate}(\text{pp}, \{\text{pk}_i\}_{i \in [N]}) \rightarrow (\text{mpk}, \{\text{hint}_i\}_{i \in [N]})$

size $\text{poly}(\lambda, \log N)$

$\text{Encrypt}(\text{mpk}, x, m) \rightarrow \text{ct}$

$\text{Decrypt}(\text{hint}_i, \text{sk}_i, x, \text{ct}) \rightarrow m$

ABE without a central authority

Prior Work

- **Registered Attribute-Based Encryption:**
 - Indistinguishability obfuscation [HLWW23]
 - Witness encryption [FWW23]
 - Pairings [HLWW23,ZZGQ23,GLWW24,AT24]: require a **structured** setup, a **bounded** number of users, and limited to **NC1 circuits**
- **Distributed Broadcast Encryption:**
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Can we do more from lattice assumptions?

Our Results (Distributed Broadcast Encryption)

- [CHW25]: **Adaptively-secure** DBE for N users from succinct LWE in the random oracle model
 - **Public parameter size:** N^2
 - **User public key size:** N
 - **Master public key, helper key, and secret key size:** $O(1)$
 - **Ciphertext size:** $O(1)$

First adaptive scheme from a falsifiable lattice assumption!
- [WW25]: Selectively-secure DBE for **unbounded** users from succinct LWE
 - **Public parameter size:** $O(1)$ transparent setup from decomposed LWE
 - **User public key size:** $O(1)$
 - **Master public key, helper key, and secret key size:** $O(1)$
 - **Ciphertext size:** $O(1)$

First scheme with unbounded users or transparent setup without iO/WE!

$\text{poly}(\lambda, \log N)$ factors omitted

Our Results (Registered ABE)

- [CHW25]: Registered ABE for input length ℓ depth d circuits and N users from succinct LWE in the random oracle model
 - **Public parameter size:** $N^2 + \ell^2$
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First scheme for circuits without iO/WE!
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First scheme with unbounded users or transparent setup without iO/WE!

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Level of efficiency only known from iO previously!

Key Challenges

1. **Functionality:** want users to generate public keys that can be *succinctly aggregated* and secret keys that facilitate correctness
2. **Security:** need to simulate a challenge ciphertext where the public keys are *adversarially chosen*

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In the lattice setting, ciphertexts often have components of the form:

$$\boxed{s^T} \boxed{M} + \boxed{e^T}$$

What if M is chosen adversarially?

Succinct LWE Family of Assumptions

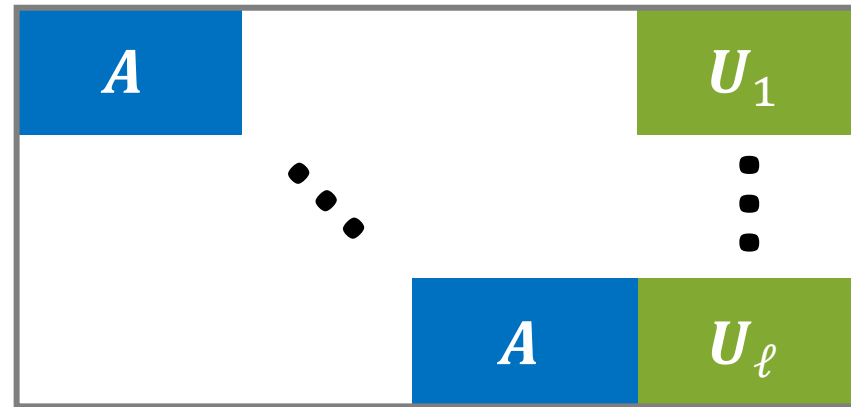
LWE is hard with respect to A given a “fresh” trapdoor for a related matrix D_ℓ :

$$\boxed{s^T} \boxed{A} + \boxed{e^T} \approx \boxed{z^T}$$

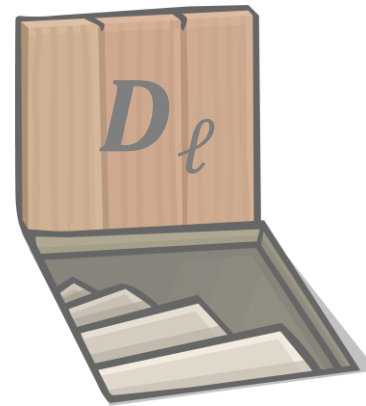
s, z are uniform, e is Gaussian

given

$D_\ell =$



and



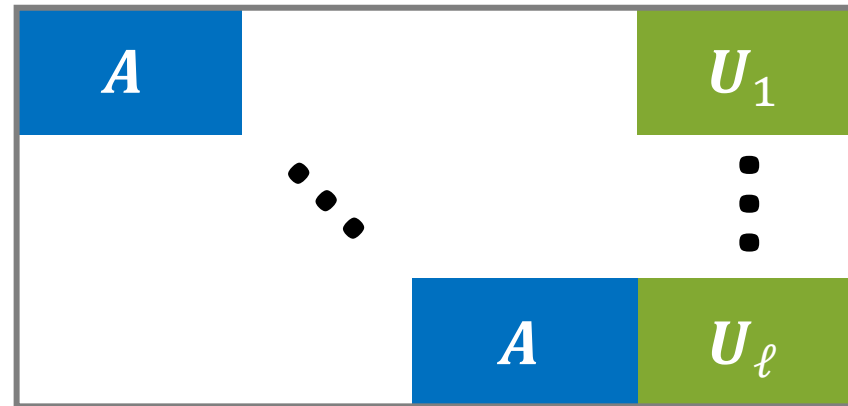
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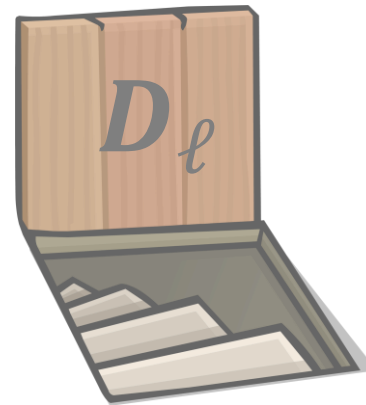
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BASIS $\Rightarrow$ Succinct $\Rightarrow$ Decomposed

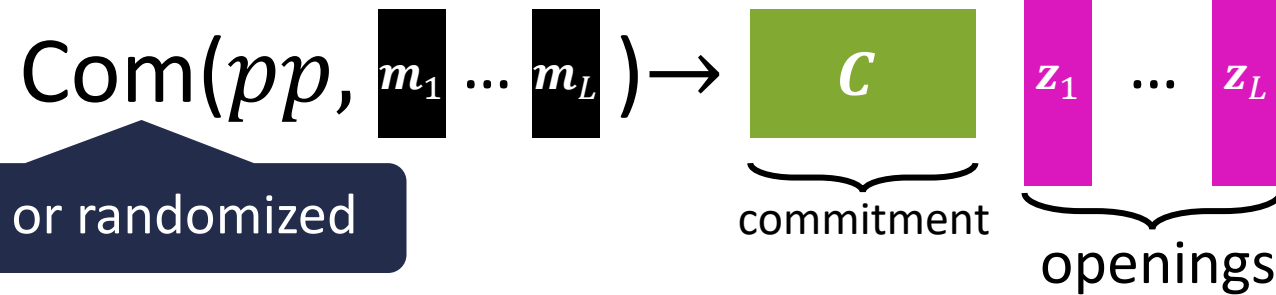
Instantiations

- BASIS [WW23]: A is uniform, U_i is structured
- Succinct LWE [Wee24]: A is uniform, U_i is uniform
- Decomposed LWE [AMR25]: A is structured, U_i is uniform, trapdoor is *public*

Matrix Commitments

[Wee25, adapted]

“vector commitment to vectors”



can be deterministic or randomized

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[Wee25, adapted]

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$$\text{Com}(pp, \mathbf{m}_1 \dots \mathbf{m}_L) \rightarrow \underbrace{\mathcal{C}}_{\text{commitment}} \underbrace{\mathbf{z}_1 \dots \mathbf{z}_L}_{\text{openings}}$$

can be deterministic or randomized

For all $i \in [L]$:

$$\mathcal{C} \mathbf{v}_i = \mathbf{m}_i - A \mathbf{z}_i$$

low-norm

pp describes A
and $\mathbf{v}_i$ for $i \in [L]$

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low-norm

pp describes $\mathbf{A}$
and $\mathbf{v}_i$ for $i \in [L]$

Security: $\mathbf{s}^T \mathbf{A} + \mathbf{e}^T \approx \mathbf{z}^T$ given pp

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Can view this as an algebraic version of the witness encryption approach in [FWW23]

DBE For Unbounded Users [WW25]

Public Parameters

pp_{com}

p

includes A, v_i

deterministic commitment

[Wee25] has fixed-size

public parameters!

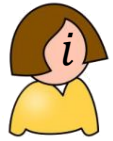
DBE For Unbounded Users [WW25]

Public Parameters

pp_{com}

p

includes A, v_i



Public Key

t_i

=

A

y_i

low-norm secret key

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Public Key

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Encryption of $b \in \{0,1\}$ to S :

Noise omitted

s^T

p

masks b

DBE For Unbounded Users [WW25]

Public Parameters

pp_{com}

p

includes A, v_i



Public Key

t_i

=

A

y_i

low-norm secret key

Encryption of $b \in \{0,1\}$ to S :

Noise omitted

s^T

C_S

commit to $p + t_j$ at position j
when $j \in S$ and 0 otherwise

can efficiently commit to exponential
number of vectors if poly many are nonzero

DBE For Unbounded Users [WW25]

Public Parameters

pp_{com}

p

includes A, v_i



Public Key

t_i

=

A

y_i

low-norm secret key

Encryption of $b \in \{0,1\}$ to S :

s^T

A

s^T

C_S

s^T

p

masks b

Noise omitted

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DBE For Unbounded Users [WW25]

Public Parameters

pp_{com}

p

includes A, v_i



Public Key

t_i

=

A

y_i

low-norm secret key

Encryption of $b \in \{0,1\}$ to S :

s^T

A

s^T

C_S

s^T

p

masks b

Noise omitted

For all $i \in S$: $C_S v_i = p + t_i - A z_i$

public and low-norm

DBE For Unbounded Users [WW25]

Public Parameters

pp_{com}

p

includes A, v_i



Public Key

t_i

=

A

y_i

low-norm secret key

Encryption of $b \in \{0,1\}$ to S :

s^T

A

s^T

C_S

s^T

p

masks b

Noise omitted

For all $i \in S$: $C_S v_i = p + t_i - A z_i$

$$s^T C_S v_i \approx s^T p + s^T t_i - s^T A z_i$$

DBE For Unbounded Users [WW25]

Public Parameters

pp_{com}

p

includes A, v_i



only user i can compute

$$s^T t_i \approx s^T A y_i$$

low-norm **secret key**

Encryption of $b \in \{0,1\}$ to S :

$$s^T A$$

$$s^T C_S$$

$$\underbrace{s^T p}_{\text{masks } b}$$

Noise omitted

$$\text{For all } i \in S: C_S v_i = p + t_i - A z_i$$

$$s^T C_S v_i \approx s^T p + s^T t_i - s^T A z_i$$

$$s^T C_S v_i + s^T A z_i \approx s^T p + s^T t_i$$

public

dual-Regev ciphertext under pk_i

Extending to Registered ABE [CHW25]

- Each user will sample a (dual-Regev) key pair and publish the public key combined with a function encoding
- Aggregation/Encryption does the following:
 1. Merkle hashes the public keys of all registered users (via the matrix commitment) and publishes the hash as the master public key
 2. Encrypts the message with respect to the hash using dual-Regev and provides a policy check that depends on the attribute
- To decrypt, registered user i can use the local opening at position i to derive an encryption of the message under their combined public key and decrypt using their secret key iff the policy check passes

Lattice-based Homomorphic Computation

[GSW13,BGGHNSVV14]

Given any input $x \in \{0,1\}^\ell$, function $f: \{0,1\}^\ell \rightarrow \{0,1\}$, and matrix B of appropriate size, there exists a **low-norm** matrix $H_{B,f,x}$ such that the following holds:

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$$\begin{array}{c} \text{encoding of } x \text{ with respect to } B \\ \boxed{B - x^T \otimes G} \end{array} \begin{array}{c} \boxed{H_{B,f,x}} \end{array} = \begin{array}{c} \boxed{B_f} \\ \text{encoding of } f(x) \text{ with respect to } B_f \end{array} - f(x) \cdot \begin{array}{c} \text{public gadget matrix} \\ \boxed{G} \end{array}$$

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$$(B - x^T \otimes G) H_{B,f,x} = B_f - f(x) \cdot G$$

Given additionally a vector d , there exists a **low-norm** vector $h_{B,f,x,d} = H_{B,f,x} G^{-1}(d)$ such that:

$$(B - x^T \otimes G) h = b_{f,d} - f(x) \cdot d$$

Registered ABE For Unbounded Users [WW25]

Public Parameters

pp_{com}

p



Public Key

t_i

$=$

A

y_i

low-norm secret key

Encryption of $b \in \{0,1\}$ to $x \in \{0,1\}^\ell$:

s^T

A

s^T

C_S

s^T

p

masks b

Noise omitted

commit to $p + t_j$ at position j
when $j \in S$ and $\mathbf{0}$ otherwise

Registered ABE For Unbounded Users [WW25]

Public Parameters

pp_{com}

p



Public Key

t_i

$=$

A

y_i

low-norm secret key

master public key

Encryption of $b \in \{0,1\}$ to $x \in \{0,1\}^\ell$:

s^T

A

s^T

C_{all}

s^T

p

masks b

Noise omitted

commit to $p + t_j$ at position $j \in [N]$

Registered ABE For Unbounded Users [WW25]

Public Parameters

pp_{com} p

B



Public Key

$t_i = A y_i$

low-norm secret key

master public key

Encryption of $b \in \{0,1\}$ to $x \in \{0,1\}^\ell$:

s^T A

s^T $B - x^T \otimes G$

s^T C_{all}

s^T p

Noise omitted

masks b

commit to $p + t_j$ at position $j \in [N]$

grows with ℓ but can be compressed via [Wee25]

Registered ABE For Unbounded Users [WW25]

can be derived from random oracle

Pub

pp_{com}

p

d_i

$i \in [N]$

B



Also includes NIZK PoK of y_i

Public Key

t_i

$=$

A

y_i

$+$

b_{f,d_i}

low-norm secret key

Encryption of $b \in \{0,1\}$ to $x \in \{0,1\}^\ell$:

s^T

A

s^T

C_{all}

s^T

p

masks b

commit to $p + t_j$ at position $j \in [N]$

Noise omitted

Registered ABE For Unbounded Users [WW25]

Public Parameters

$$pp_{\text{com}} \quad p \quad d_i$$

$$B \quad i \in [N]$$



Public Key

$$t_i = A \quad y_i + b_{f,d_i}$$

low-norm secret key

Encryption of $b \in \{0,1\}$ to $x \in \{0,1\}^\ell$:

$$s^T \quad A \quad s^T \quad C_{\text{all}} \quad s^T \quad p$$

$$s^T \quad B - x^T \otimes G$$

commit to $p + t_j$ at position $j \in [N]$

noise omitted

masks b

Recall:

$$B - x^T \otimes G \quad h = b_{f,d} - f(x) \cdot d$$

$h = h_{B,f,x,d}$ is low-norm

Registered ABE For Unbounded Users [WW25]

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commit to $p + t_j$ at position $j \in [N]$

noise omitted

masks b

$$s^T t_i \approx s^T A y_i + s^T (B - x^T \otimes G) h$$

if and only if $f_i(x) = 0$

Public Key

$$t_i = A y_i + b_{f,d_i}$$

low-norm secret key

Recall:

$$B - x^T \otimes G \quad h = b_{f,d} - f(x) \cdot d$$

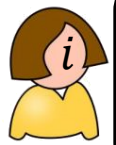
$h = h_{B,f,x,d}$ is low-norm

Registered ABE For Unbounded Users [WW25]

Public Parameters

$$pp_{\text{com}} \quad p \quad d_i$$

$$B \quad i \in [N]$$



$$s^T t_i \approx s^T A y_i + s^T (B - x^T \otimes G) h$$

if and only if $f_i(x) = 0$

Public Key

$$t_i = A y_i + b_{f,d_i}$$

low-norm secret key

Encryption of $b \in \{0,1\}$ to $x \in \{0,1\}^\ell$:

$$s^T A \quad s^T C_{\text{all}} \quad \underbrace{s^T p}_{\text{masks } b}$$

$$s^T (B - x^T \otimes G)$$

Noise omitted

$$\text{For all } i \in [N]: C_{\text{all}} v_i = p + t_i - A z_i$$

$$s^T C_{\text{all}} v_i \approx s^T p + s^T t_i - s^T A z_i$$

$$\underbrace{s^T C_{\text{all}} v_i + s^T A z_i}_{\text{public}} \approx \underbrace{s^T p + s^T t_i}_{\text{dual-Regev ciphertext under } pk_i}$$

public

dual-Regev ciphertext under pk_i

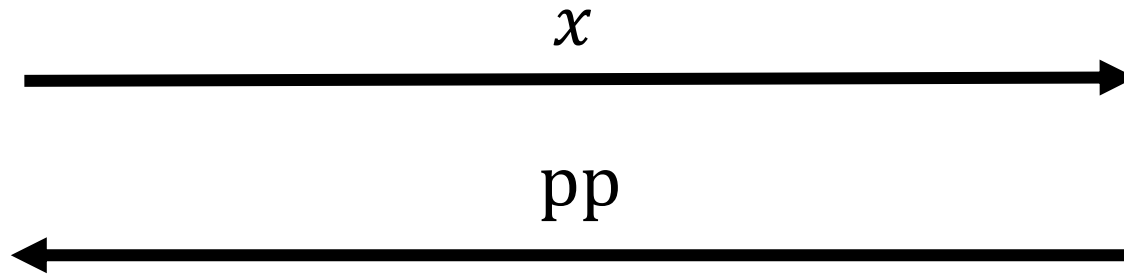
Registered ABE (Selective) Security

[HLWW23]

Adversary



Challenger



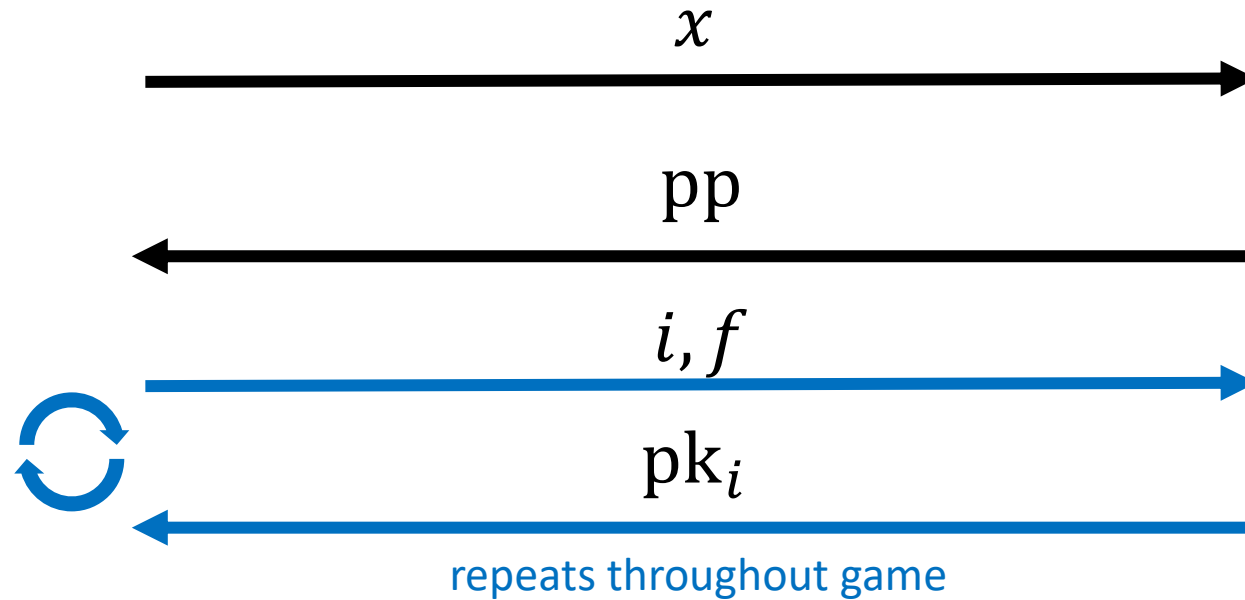
Registered ABE (Selective) Security

[HLWW23]

Adversary



Challenger



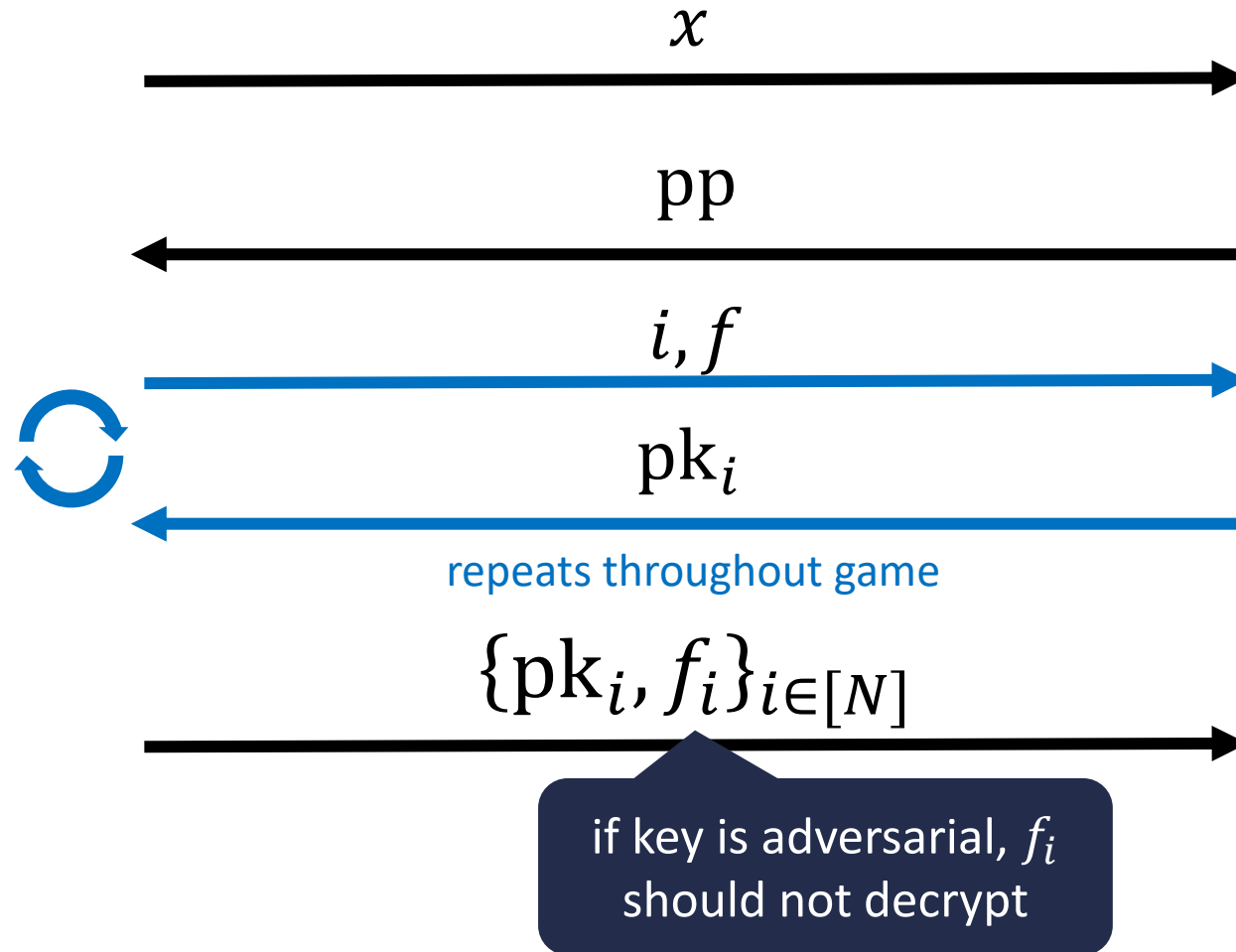
Registered ABE (Selective) Security

[HLWW23]

Adversary



Challenger



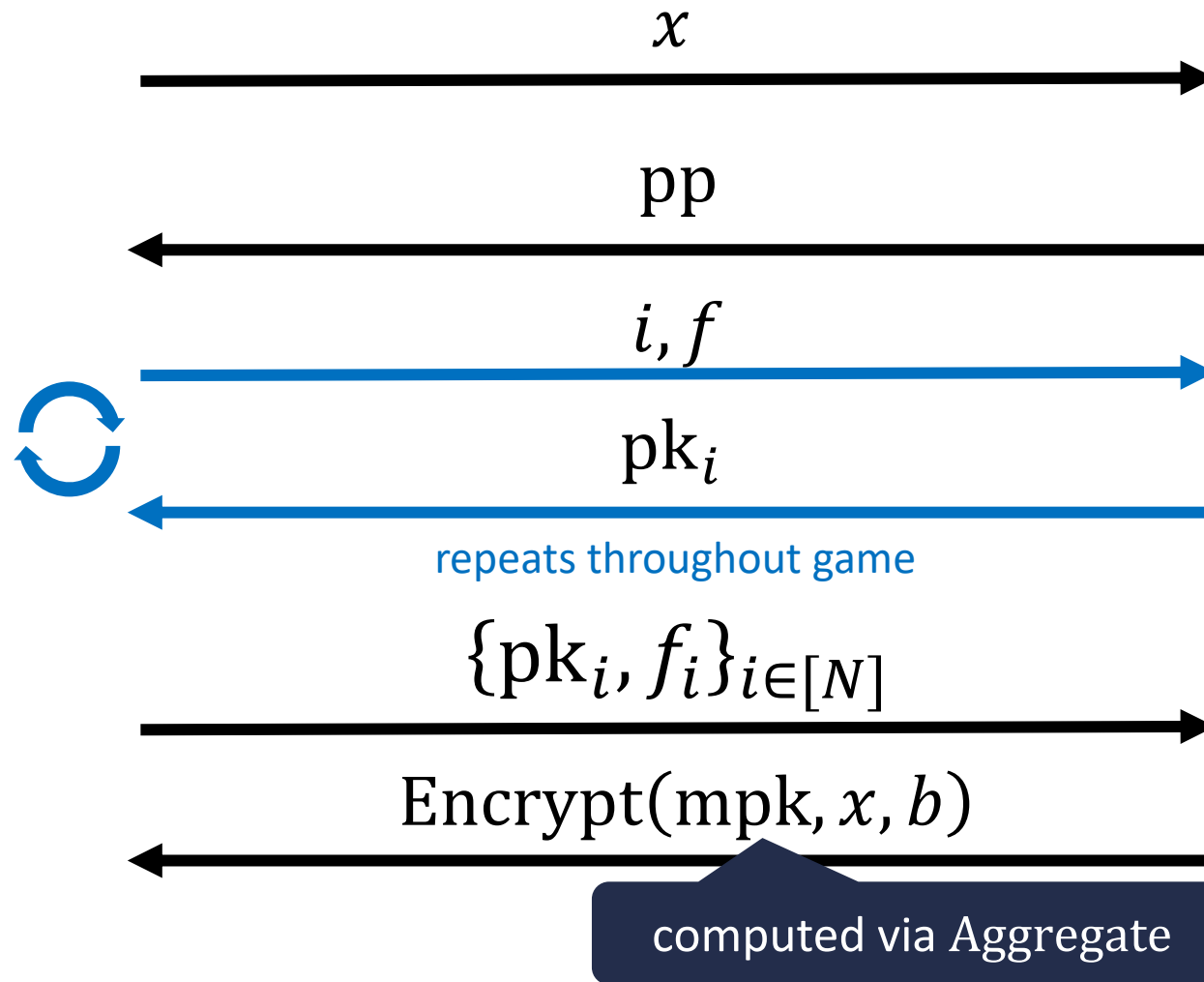
Registered ABE (Selective) Security

[HLWW23]

Adversary



Challenger



challenge bit $b \in \{0,1\}$
sampled randomly

Proving Security [CHW25]

Public Parameters

$$pp_{\text{com}} \quad p \quad d_i$$

B $i \in [N]$



Public Key

$$t_i = A y_i + b_{f,d_i}$$

low-norm secret key

Encryption of $b \in \{0,1\}$ to $x \in \{0,1\}^\ell$:

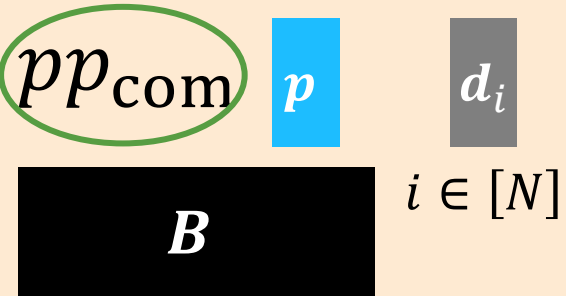
$$s^T A \quad s^T C_{\text{all}} \quad s^T p$$

$s^T B - x^T \otimes G$ commit to $p + t_j$ at position $j \in [N]$

Noise omitted

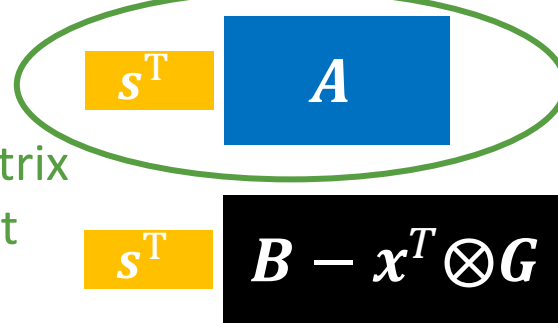
Proving Security [CHW25]

Public Parameters



given by matrix
commitment
security

Encryption of $b \in \{0,1\}$ to $x \in \{0,1\}^\ell$:



commit to $p + t_j$ at position $j \in [N]$

Noise omitted



Public Key

$$t_i = A y_i + b_{f,d_i}$$

low-norm secret key

Proving Security [CHW25]

Public Parameters

pp_{com}

p

d_i

$i \in [N]$

B

given by matrix
commitment
security

Encryption of $b \in \{0,1\}$ to $x \in \{0,1\}^\ell$:

s^T

A

s^T

C_{all}

s^T

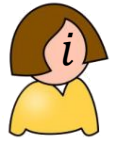
p

Noise omitted

s^T

$B - x^T \otimes G$

can set $p = Ak$ for low-norm k and
 $B = AK + x^T \otimes G$ for low-norm K



Public Key

t_i

$=$

A

y_i

$+$

b_{f,d_i}

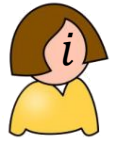
low-norm secret key

Proving Security [CHW25]

Public Parameters

$$pp_{\text{com}} \quad p \quad d_i$$

$$B \quad i \in [N]$$



Public Key

$$t_i = A y_i + b_{f,d_i}$$

low-norm secret key

Encryption of $b \in \{0,1\}$ to $x \in \{0,1\}^\ell$:

$$s^T A$$

$$s^T (B - x^T \otimes G)$$

$$s^T C_{all}$$

$$s^T p$$

Noise omitted

commits vectors t_j chosen by the adversary!

Proving Security [CHW25]

Public Parameters

$$pp_{\text{com}} \quad p \quad d_i$$

$$B \quad i \in [N]$$



Public Key

$$t_i = A y_i + b_{f,d_i}$$

low-norm secret key

Encryption of $b \in \{0,1\}$ to $x \in \{0,1\}^\ell$:

$$s^T A \quad s^T \left(C_{all} + C_0 \right) \quad s^T p$$

$$s^T \left(B - x^T \otimes G \right)$$

Noise omitted

Key idea: add a randomized commitment to $\mathbf{0}$ at encryption time to remove adversarial control

Proving Security [CHW25]

Public Parameters

$$pp_{\text{com}} \quad p \quad d_i \quad i \in [N]$$
$$B$$



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Noise omitted

Key idea: add a randomized commitment to 0 at encryption time to remove adversarial control

Problem: must provide openings to ensure correctness!

Proving Security [CHW25]

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$$pp_{\text{com}} \quad p \quad d_i$$
$$B \quad i \in [N]$$



Public Key

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Noise omitted

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Problem: must provide openings to ensure correctness!

Solution: random oracle derives C_0 and $z_{0,i}$ from aggregation query

Proving Security [CHW25]

Public Parameters

$$pp_{\text{com}} \quad p \quad d_i$$

$$B \quad i \in [N]$$



Public Key

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$$s^T A \quad s^T \left(C_{all} + C_0 \right) \quad s^T p$$

Noise omitted

$$s^T (B - x^T \otimes G)$$

RO derives randomized commitment C_0
and openings $z_{0,i}$ from aggregation query

Additional proof notes:

- Randomness of C_0 is **public** so commitment must be *explainable* [AWY20,LW22] to program $C_0 = AK_0 - C_{all}$ indistinguishably

Proving Security [CHW25]

Public Parameters

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Noise omitted

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- Randomness of C_0 is public so commitment must be *explainable* [AWY20,LW22] to program $C_0 = AK_0 - C_{\text{all}}$ indistinguishably
- Openings $z_{0,i}$ must be sufficiently high-norm; idea is to find a low-norm “opening” $z'_{0,i}$ such that $C_0 v_i = d_i - A z'_{0,i}$, program $d_i = A k_i$ where $\|k_i\| = \|z_{0,i}\|$, and set $z_{0,i} = z'_{0,i} - k_i$ as the opening to 0

Proving Security [CHW25]

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Techniques also work for adaptively-secure DBE in the ROM

Open Problems

- Proving security from plain LWE
- Removing the random oracle from registered ABE or adaptively-secure DBE
- Cryptanalysis and more applications of succinct/decomposed LWE

Thanks for listening!

<https://eprint.iacr.org/2025/044>

<https://eprint.iacr.org/2025/1039>